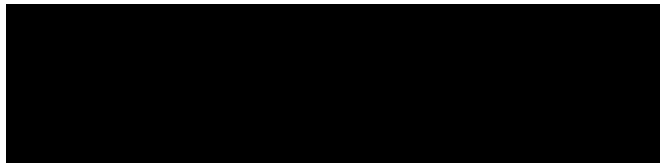
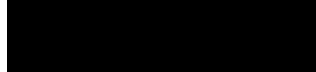


F20FO Digital Forensics Coursework 1

A Lab Analysis of Forensic Tools

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Introduction

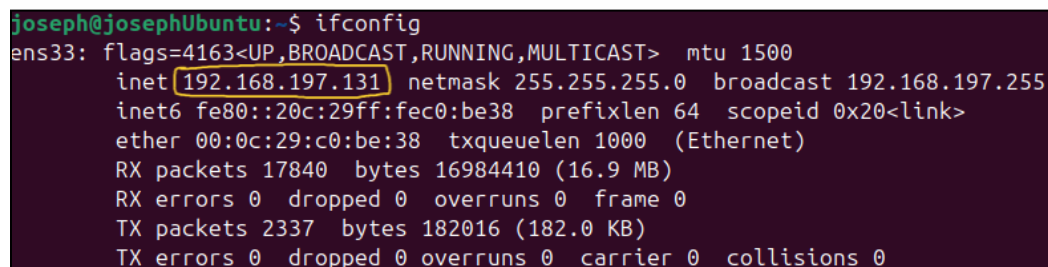
This report outlines the development and evaluation of a virtual forensics lab designed to test digital forensic tools across two virtual machines, one running Windows and the other Linux. Each machine is equipped with two specialised forensic tools, resulting in a total of four tools used in this report. Specifically, the Linux machine is configured with Autopsy and Volatility, while the Windows machine utilises NetworkMiner and OSForensics.

Lab-Setup

The virtual lab was created using VMware Workstation Pro, where both Ubuntu and Windows operating systems were installed on separate virtual machines (VMs). The VMs are configured with NAT (Network Address Translation) which essentially allows them to share the host computer's IP address for internet access. This Setup has two main benefits:

- Internal Communication: The NAT configuration creates a private virtual network that enables the two VMs to communicate with each other securely. This is crucial especially when dealing with forensic data or evidence that needs to be exchanged between machines.
- External Access: The NAT configuration also allows the VMs to access external resources on the internet, which might be needed to keep the operating system updated, or to access additional tools without compromising the isolation of the internal network.

Below are screenshots of the NAT configuration, which display the IP addresses assigned to each VM and confirm connectivity via successful ping tests.



```
joseph@josephUbuntu:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.197.131  netmask 255.255.255.0  broadcast 192.168.197.255
    inet6 fe80::20c:29ff:fec0:be38  prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:c0:be:38  txqueuelen 1000  (Ethernet)
    RX packets 17840  bytes 16984410 (16.9 MB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 2337  bytes 182016 (182.0 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

Fig 1: Ubuntu IP configurations

```
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::766:adcc:2ad3:b768%2
    IPv4 Address. . . . . : 192.168.197.129
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.197.2
```

Fig 1.1: Windows IP configurations

```
joseph@josephUbuntu:~$ ping 192.168.197.129
PING 192.168.197.129 (192.168.197.129) 56(84) bytes of data:
64 bytes from 192.168.197.129: icmp_seq=1 ttl=128 time=0.879 ms
64 bytes from 192.168.197.129: icmp_seq=2 ttl=128 time=0.986 ms
64 bytes from 192.168.197.129: icmp_seq=3 ttl=128 time=0.910 ms
```

Fig 1.2: Ubuntu ping to Windows machine

```
C:\Users\Joseph>ping 192.168.197.131

Pinging 192.168.197.131 with 32 bytes of data:
Reply from 192.168.197.131: bytes=32 time<1ms TTL=64
Reply from 192.168.197.131: bytes=32 time<1ms TTL=64
Reply from 192.168.197.131: bytes=32 time<1ms TTL=64
Reply from 192.168.197.131: bytes=32 time=1ms TTL=64
```

Fig 1.3: Windows ping to Ubuntu machine

Forensic Tools

In this section, the report introduces all four digital forensic tools used in the lab setup. On the Linux side, Autopsy and Volatility were installed, while on the Windows side, NetworkMiner and OSForensics were installed. While the report will later provide a detailed analysis of Autopsy and NetworkMiner (The two selected tools for detailed analysis), this section outlines the functions of the additional tools selected in this forensic investigation.

Volatility

Volatility is an open-source memory forensics framework designed for analyzing volatile memory (RAM) from computer systems. It enables digital forensic investigators to extract and scrutinize

critical information such as running processes, open network connections, registry keys, file handles, and other system artifacts from a system's memory (Ltd, 2023). Additionally, Volatility provides capabilities for detecting and analyzing malware by examining the memory state for suspicious or malicious activity, making it an invaluable tool for reconstructing a system's state during an investigation.

Main Features of Volatility -:

- Analyzes volatile memory (RAM) to extract detailed system information
- Extracts registry keys, file handles, and other memory-resident data
- Capable of detecting malware by identifying suspicious memory patterns
- Can assist in constructing a timeline of events by extracting memory artifacts like process creation times and network activities

OSForensics

OSForensics allows users to identify suspicious files and activity with hash matching, drive signature comparisons, email, memory, and binary data. This software lets users extract forensic evidence from computers with advanced file searching and indexing and enables this data to be managed effectively (Va.gov, 2022). In addition to advanced file searching and indexing, OSForensics provides a disk imaging and live system monitoring option which helps capture a detailed snapshot of the target system. Combining this with its intuitive interface and robust reporting capabilities makes it a powerful tool in digital forensic investigations.

Main Features of OSForensics -:

- Identifies suspicious files and activities using hash-matching
- Offers advanced file searching and indexing capabilities
- Facilitates comprehensive disk imaging for evidence preservation
- Analyzes email content, memory, and binary data for forensic evidence

NetworkMiner

NetworkMiner is an open-source Network Forensic Analysis Tool (NFAT) primarily for Windows, though it can also be used on Linux. It passively captures network packets to detect operating systems, sessions, hostnames, open ports, and more without generating additional network traffic (Ltd, 2023). Additionally, NetworkMiner can reassemble packets for analyzing unencrypted traffic and identify network anomalies by monitoring port loads, source, and destination addresses.

Hosts Tab

The picture below shows the host tab, which displays all the information about each captured host when the pcap file was analysed. Details such as IP address, hostname, operating system

used, total packets sent and received, and even all the outgoing connections from the host are all displayed in a clear tree view.

Below is information about a host with IP `172.16.1.66` with other information such as the MAC address, the NIC vendor, and the OS associated with the host.

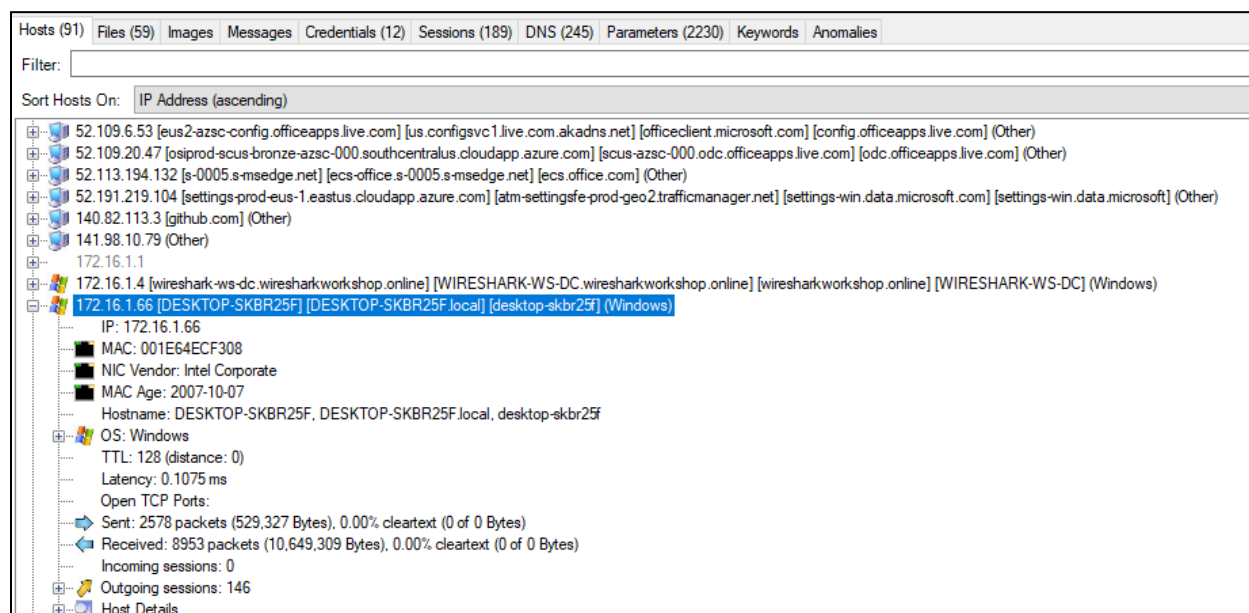


Fig 2: Host Information

The host tab seems to have a great user interface as it presents complex network traffic data in a clean and organised manner, which is important when analyzing complex data. The tool also uses passive fingerprinting techniques to guess each host's operating system, which is also important as identifying which OS the user had used can be forensically relevant (Jasper, 2025). However, due to the limited available filtering options, it could be hard to sift through a huge PCAP file with lots of hosts and could present a problem.

File View and Extraction

The file tab automatically reassembles files based on common protocols like HTTP, FTP, and more. This made looking through files from a PCAP much simpler and efficient compared to tools like Wireshark. Information such as timestamps and source and destination port were also labelled with the file, which could be important information when analysing. The filter option was also very helpful as filtering to find a particular file format like ".exe" was possible.

However, certain limitations were noticed when using this feature, such as the number of protocols that were supported for this feature, which were only a handful. This means if a file were to come in an unsupported format, it wouldn't be reassembled (Netresec, 2019). Another issue is performance, as the files got bigger, the performance had degraded significantly, which made the experience much worse than it should be.

Hosts (91) Files (59) Images Messages Credentials (12) Sessions (189) DNS (245) Parameters (2230) Keywords Anomalies										
Filter keyword:										
☐ Case sensitive	☐ ExactPhrase	☐ Any column	Clear							
Frame nr.	Filename	Extension	Size	Source host	S. port	Destination host	D. port	Protocol	Timestamp	
1343	Microsoft Azure RSA TLS Issu.cer	cer	1 456 B	52.109.6.53 [eus2-azsc-config.officeapps.live.com] [us.con...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49735	TlsCertificate	2024-07-30 02:39:17 UTC	
1425	odc.officeapps.live.com.cer	cer	2 476 B	52.109.0.142 [oiprod-wus-bronze-azsc-000.westus.clouda...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49737	TlsCertificate	2024-07-30 02:39:18 UTC	
1425	Microsoft Azure RSA TLS Issu.cer	cer	1 456 B	52.109.0.142 [oiprod-wus-bronze-azsc-000.westus.clouda...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49737	TlsCertificate	2024-07-30 02:39:18 UTC	
1466	odc.officeapps.live.com[1].cer	cer	2 476 B	52.109.0.142 [oiprod-wus-bronze-azsc-000.westus.clouda...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49738	TlsCertificate	2024-07-30 02:39:23 UTC	
1466	Microsoft Azure RSA TLS Issu[1].cer	cer	1 456 B	52.109.0.142 [oiprod-wus-bronze-azsc-000.westus.clouda...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49738	TlsCertificate	2024-07-30 02:39:23 UTC	
1475	g.msn.com.cer	cer	2 521 B	20.241.44.114 [g-msn-com-rsatic.trafficmanager.net] [g.ms...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49739	TlsCertificate	2024-07-30 02:39:23 UTC	
1475	Microsoft Azure RSA TLS Issu.cer	cer	1 456 B	20.241.44.114 [g-msn-com-rsatic.trafficmanager.net] [g.ms...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49739	TlsCertificate	2024-07-30 02:39:23 UTC	
1506	sfx.ms.cer	cer	2 154 B	23.52.9.140 [e9659.dapp.akamaiedge.net] [oneclient.sfx.m...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49740	TlsCertificate	2024-07-30 02:39:24 UTC	
1506	Microsoft Azure RSA TLS Issu.cer	cer	1 456 B	23.52.9.140 [e9659.dapp.akamaiedge.net] [oneclient.sfx.m...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49740	TlsCertificate	2024-07-30 02:39:24 UTC	
1506	DigCert Global Root G2.cer	cer	914 B	23.52.9.140 [e9659.dapp.akamaiedge.net] [oneclient.sfx.m...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49740	TlsCertificate	2024-07-30 02:39:24 UTC	

Fig 2.1: File Information

Figure 2.4 shows a suspicious connection made to *repo1.maven.org*

1836	github.com.cer	cer	1 191 B	140.82.113.3 [github.com]	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49749	TlsCertificate	2024-07-30 02:39:53 UTC	
1836	Sectigo ECC Domain Validatio.cer	cer	940 B	140.82.113.3 [github.com]	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49749	TlsCertificate	2024-07-30 02:39:53 UTC	
1836	USERTrust ECC Certification.cer	cer	983 B	140.82.113.3 [github.com]	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49749	TlsCertificate	2024-07-30 02:39:53 UTC	
1866	github.io.cer	cer	1 853 B	185.199.110.133 [objects.githubusercontent.com]	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49750	TlsCertificate	2024-07-30 02:39:53 UTC	
1866	DigCert Global G2 TLS RSA S.cer	cer	1 228 B	185.199.110.133 [objects.githubusercontent.com]	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49750	TlsCertificate	2024-07-30 02:39:53 UTC	
2027	repo1.maven.org.cer	cer	1 637 B	199.232.196.209 [dualstack.sonatype.map.fastly.net] [repo...	TCP 443	172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	TCP 49751	TlsCertificate	2024-07-30 02:39:57 UTC	

Fig 2.2: File Information (2)

Figure 2.5 shows the certificate after extracted from the file, which seems to be invalid or expired

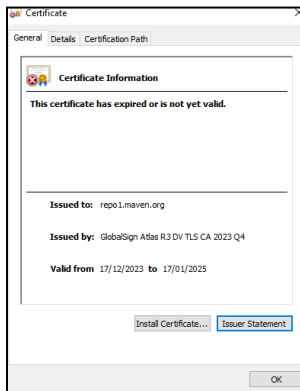


Fig 2.3: Invalid Certificate

Credentials View and Extraction

The credential tab within the tool showcases all the credentials that were transmitted within the network traffic. Usually the tool captures credentials in clear text, but for this case, it captured the hashes for the Kerberos authentication, which is also valuable because if the password is weak, then the hash can be cracked a little easier, which could reveal even more information (Netresec, 2019b). The credentials tab is presented in a clear, user-friendly way, which is also easily readable. There are also options to hide/show cookies, NTLM handshakes, and passwords for preference when dealing with this type of data.

However, this tool becomes not so useful if the data is encrypted in transit and only the hash of the password is captured. Trying to break the hash would present a challenge on its own and would also take time. While this tool does focus on credential pattern, there is a possibility of misidentification of data where the tool might interpret something as credentials when it's not and could cause confusion (Hjelmvik, 2019).

Hosts (91) Files (59) Images Messages Credentials (12) Sessions (189) DNS (245) Parameters (230) Keywords Anomalies									
☒ Show Cookies ☒ Show NTLM challenge-response ☐ Mask Passwords									
Client	Server	Protocol	Username	Password	Valid login	First Login			
[172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	ccollier	\$krb5pa\$18\$ccollier\$WIRESHARKWORKSH\$WIRESHA...	Unknown	2024-07-30 02:39:12 UTC			
172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	ccollier	\$krb5seep\$18\$WIRESHARKWORKSHOP.ONLINEccolli...	Unknown	2024-07-30 02:39:12 UTC			
172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	WIRESHARKWORKSHOP.ONLINEhostdesktop-ekbr25f...	\$krb5seep\$18\$WIRESHARKWORKSHOP.ONLINEhostd...	Unknown	2024-07-30 02:38:49 UTC			
172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	WIRESHARKWORKSHOP.ONLINEhostdesktop-ekbr25f...	\$krb5seep\$18\$WIRESHARKWORKSHOP.ONLINEhostd...	Unknown	2024-07-30 02:38:49 UTC			
172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	WIRESHARKWORKSHOP.ONLINEhostdesktop-ekbr25f...	\$krb5pa\$18\$wiresharkworkshop.online\$WIRESHARKW...	Unknown	2024-07-30 02:38:49 UTC			
172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	WIRESHARKWORKSHOP.ONLINEhostdesktop-ekbr25f...	\$krb5seep\$18\$WIRESHARKWORKSHOP.ONLINEhostd...	Unknown	2024-07-30 02:38:49 UTC			

Fig 2.4: Credential Information

Figure 2.7 shows a username associated with an authentication

[172.16.1.66 [DESKTOP-SKBR25F] [DESKTOP-SKBR25F]...	172.16.1.4 [wireshark-ws-dc.wiresharkworkshop.online] [W...	Kerberos	ccollier
---	---	----------	----------

Fig 2.5: Username Found

Figure 2.8 shows the password hash associated with the username

\$krb5pa\$18\$ccollier\$WIRESHARKWORKSH\$WIRESHARKWORKSHOP.ONLINEccollier\$dfc97801ec8f2f862f8aeb03ec2dfbe794178cfaaddeebef2076b4f82dde91a5f5f8ad9ced06df948870278e81f83453422413c475dbd01
--

Fig 2.6: Password Hash

Autopsy

Autopsy is an open-source digital forensics platform that provides a GUI interface for analyzing digital media. It integrates a wide range of forensic tools, including those from The Sleuth Kit, to help investigators recover and examine data from hard drives, smartphones, and other storage devices. Its features include timeline analysis, keyword searching, file system analysis, and artifact extraction, making it a powerful solution for both law enforcement and forensic investigations (Autopsy, 2023).

Directory View and File Information

When a disk image is loaded into Autopsy, the tool automatically generates a structured directory view that simplifies navigation through the file system. This interface enables an efficient browsing of files, which is crucial when trying to visualize complex data. The directory also shows all files within the disk, including those hidden and deleted.

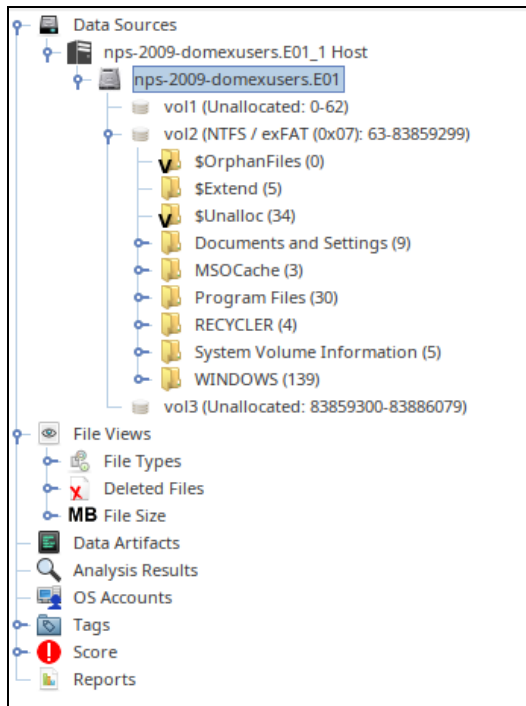


Fig 3: Directory Tree

Figure 3.3 shows an extracted image file from a folder

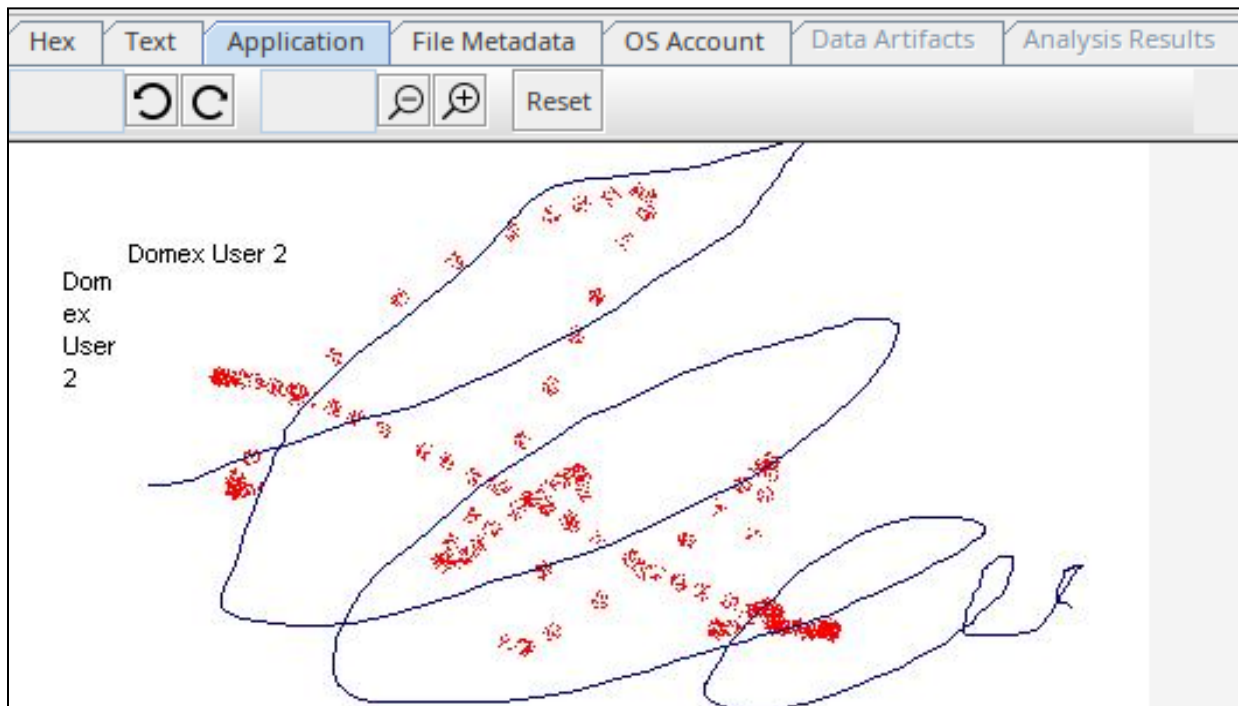


Fig 3.1: Image Extracted

Actions such as extraction and metadata viewing of all files are also possible within the directory, which is very beneficial when trying to examine certain files in more depth. However, this organization of large data into a structural way of navigation can turn out to be difficult, especially in very big disks, as the tool doesn't highlight any anomalies and manual search is the only way.

Another problem that seemed to be persistent is the performance degradation on Large disks, as trying to navigate or even open a file becomes sluggish (Virtualcyberlabs.com, 2025).

Figure 3.6 shows the extracted files metadata for further analysis..

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata									
Name:	/img_nps-2009-domexusers.E01/vol_vol2/Documents and Settings/domex2/My Documents/My Pictures/domexuser2.JPG								
Type:	File System								
MIME Type:	image/jpeg								
Size:	19920								
File Name Allocation:	Allocated								
Metadata Allocation:	Allocated								
Modified:	2008-10-29 20:21:16 GST								
Accessed:	2008-10-30 05:51:26 GST								
Created:	2008-10-29 20:21:16 GST								
Changed:	2008-10-30 05:59:10 GST								
MD5:	Not calculated								
SHA-256:	Not calculated								
Hash Lookup Results:	UNKNOWN								

Fig 3.2: Image Metadata

Timeline

The Autopsy timeline feature gives a clear picture of when files were created, modified, and accessed based on metadata. All of this information is put into an interactive timeline, which can be accessed by simply clicking the year that needs to be seen. This makes it easier to identify patterns and pinpoint critical moments in the data (1). The zoom feature present within the timeline makes it an even better tool as certain points in time or certain events can be magnified and shown in more detail (2). Timeline also can be filtered to show based on certain file types or even certain specific metadata (3).

However, even with all these additional features, it still seemed like a complex tool to use, nevertheless and combine this with the performance limitation with a big disk, it's not the best user experience (Virtualcyberlabs.com, 2025).

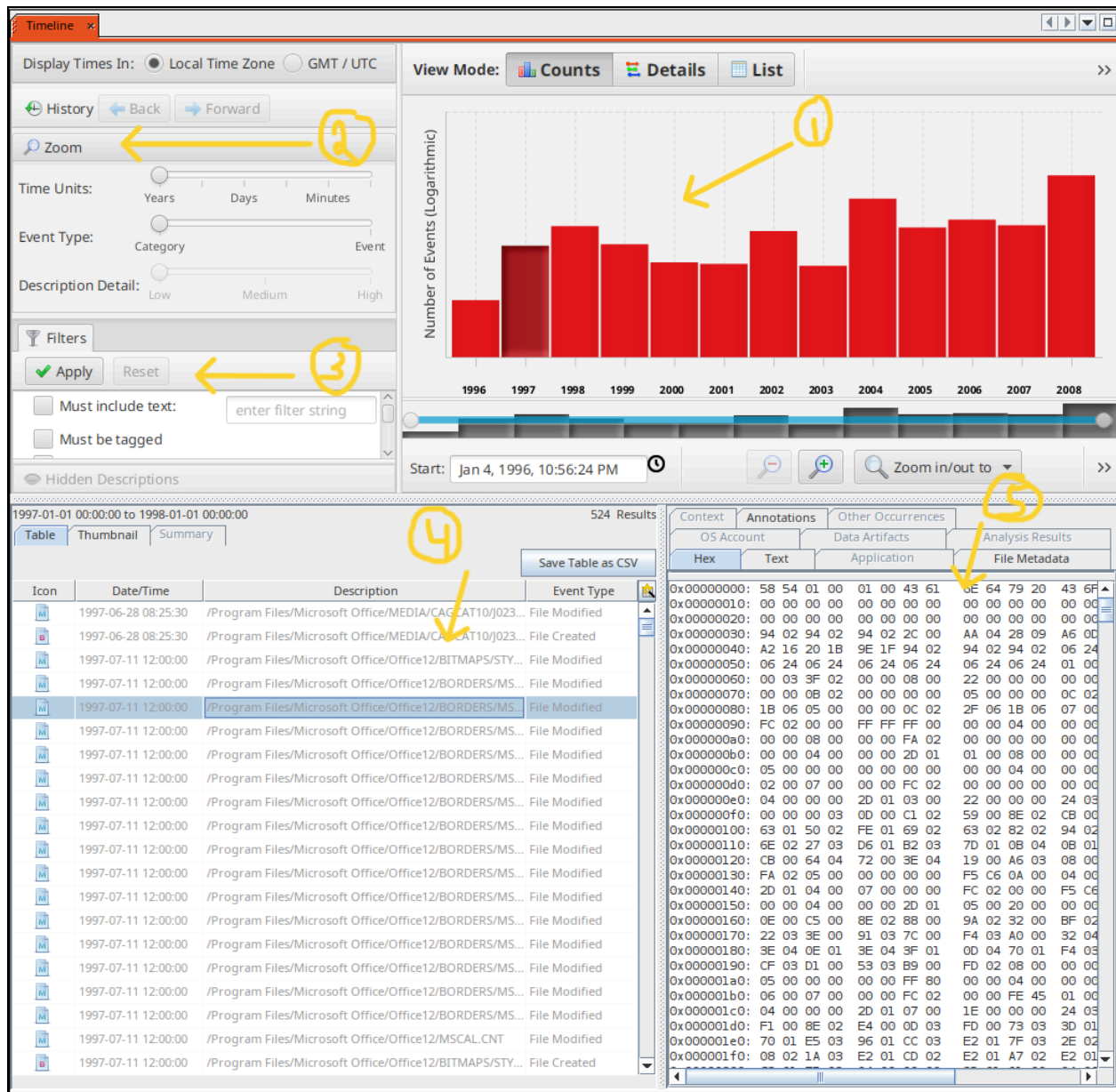
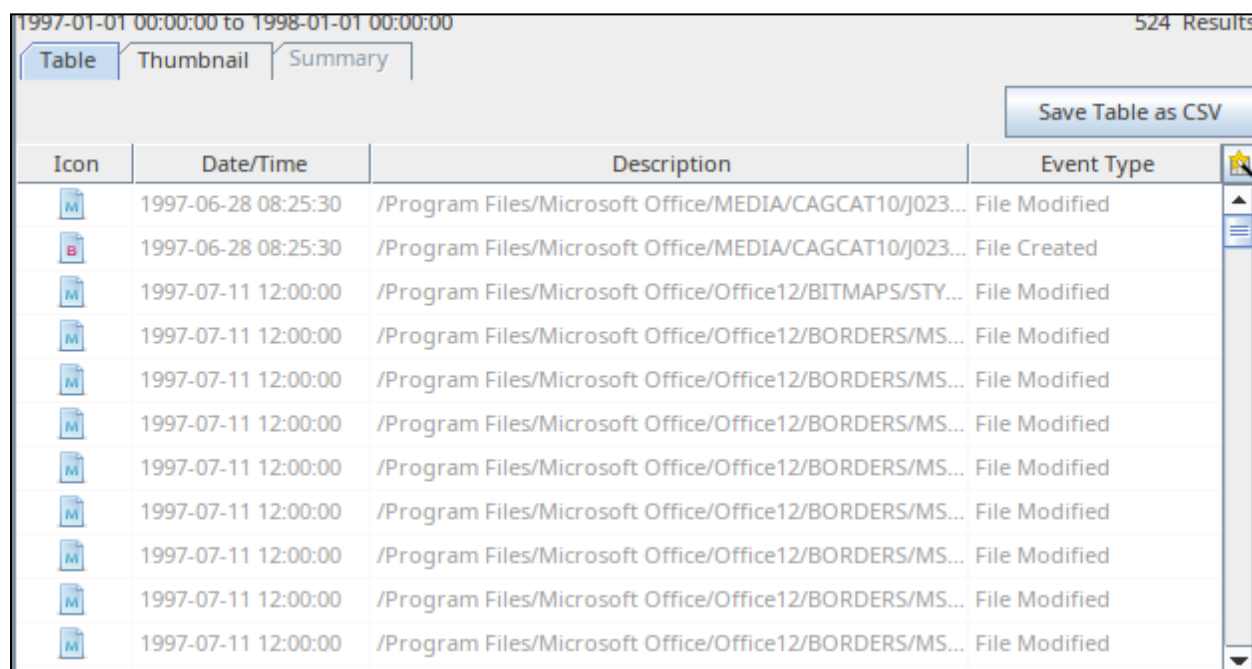


Fig 3.3: Timeline Viewer

Right bottom of the image, there is the hexdump for the particular file selected from the timeline (5).

Figure 3.6 shows a list of files created and modified for the year 1997 within the timeline (4).



Icon	Date/Time	Description	Event Type
	1997-06-28 08:25:30	/Program Files/Microsoft Office/MEDIA/CAGCAT10/J023...	File Modified
	1997-06-28 08:25:30	/Program Files/Microsoft Office/MEDIA/CAGCAT10/J023...	File Created
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BITMAPS/STY...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified
	1997-07-11 12:00:00	/Program Files/Microsoft Office/Office12/BORDERS/MS...	File Modified

Fig 3.4: Timeline Viewer (Files 1997)

Report Generation

Autopsy also has a report generation function where a user can tag certain evidence or important information they think is vital to the case they are solving and can generate a report based on those tagged results. This narrows down the items within the disk to what's critical to the case, which makes this streamlined and efficient. The report can also be generated in different formats such as HTML, XML, and more, which is great if someone has a preference.

However, even when the report generation is user-friendly and very easy to use, there seems to be a lack of deep customization specifically to generate a report in a more detailed way, which could be bad in cases where the detail is the answer to an investigation. Another issue that had been persistent is the report generation for a large amount of tagged items, which took a long time to generate and had sluggish performance when used.

Figure 3.10 shows a report generated with the HTML template using Autopsy. It was generated using one tagged file.

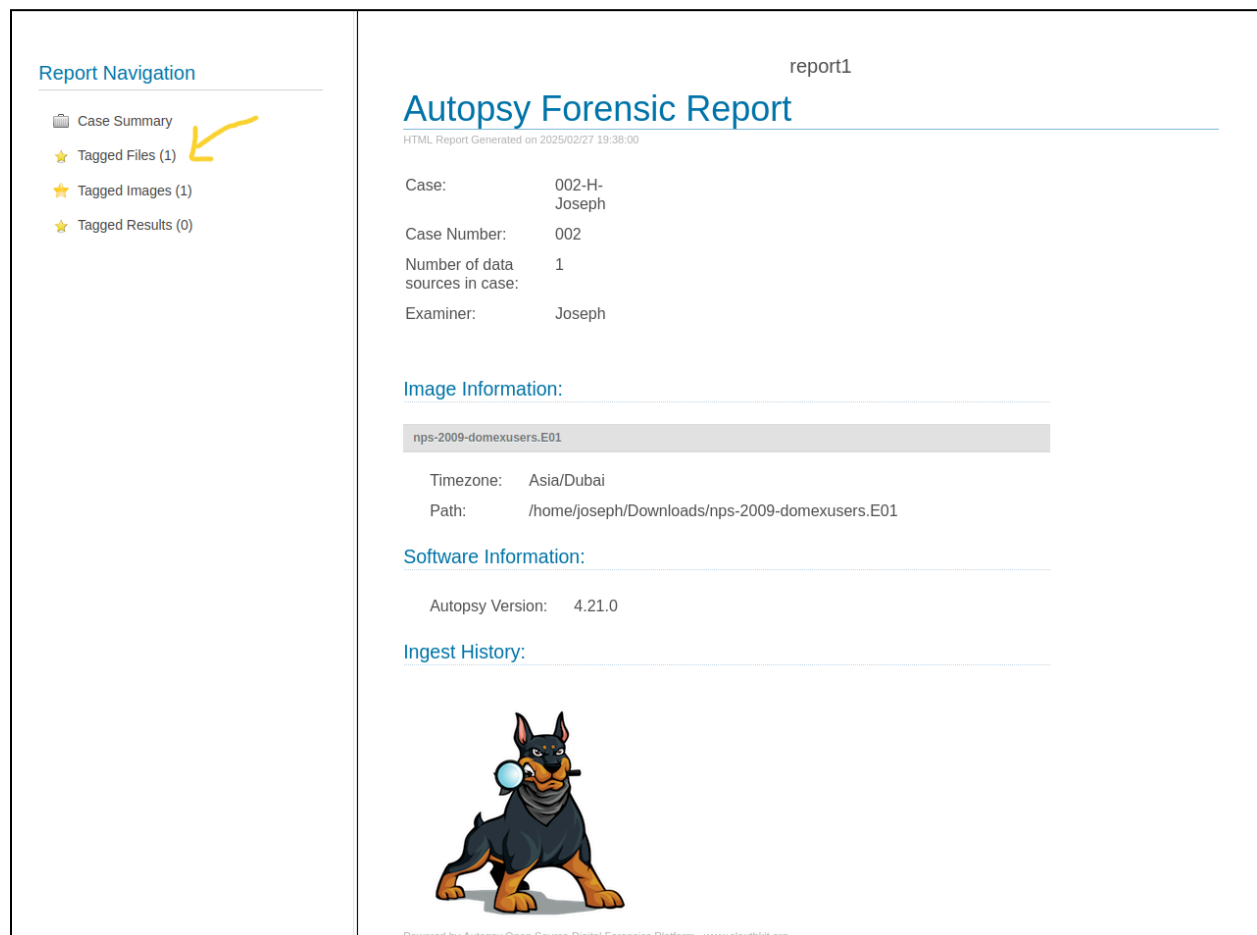


Fig 3.5: Report Generation (Open)

Conclusion

Overall, the usage of these tools seemed to be both effective and very insightful, not only do they streamline the forensic process by automating extractions of key artifacts such as files and credentials, but also offer a highly intuitive and user-friendly interface for in-depth analysis. Despite having some flaws particularly when handling large datasets, these tools do provide

valuable and actionable information that could enhance an investigation. Overall a great assessment and a great learning experience.

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Appendices

Image info from Volatility

```
joseph@josephUbuntu:~/Documents$ volatility -f Snapshot6_1609157562389.vmem imageinfo
Volatility Foundation Volatility Framework 2.6.1
INFO      : volatility.debug      : Determining profile based on KDBG search...
           Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_24000, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP1x64_24000, Win7SP1x64_23418
           AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
           AS Layer2 : FileAddressSpace (/home/joseph/Documents/Snapshot6_1609157562389.vmem)
           PAE type  : No PAE
           DTB       : 0x187000L
           KDBG      : 0xf80002c4a0a0L
           Number of Processors : 1
           Image Type (Service Pack) : 1
           KPCR for CPU 0 : 0xffffffff80002c4bd00L
           KUSER_SHARED_DATA : 0xffffffff78000000000L
           Image date and time : 2020-12-27 06:20:05 UTC+0000
           Image local date and time : 2020-12-26 22:20:05 -0800
```

Running services within the memory dump

```
joseph@josephUbuntu: ~/Documents$ volatility -f Snapshot6_1609157562389.vmem --profile Win7SP1x64 pslist
Volatility Foundation Volatility Framework 2.6.1
```

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start	Exit
0xffffffff80024b3890	System	4	0	83	519	-----	0	2020-12-27 06:18:43 UTC+0000	
0xffffffff8002c33b30	smss.exe	244	4	2	29	-----	0	2020-12-27 06:18:43 UTC+0000	
0xffffffff8002e79b30	csrss.exe	332	324	9	562	0	0	2020-12-27 06:18:49 UTC+0000	
0xffffffff8003a0f8a0	csrss.exe	384	376	10	171	1	0	2020-12-27 06:18:50 UTC+0000	
0xffffffff8003a12060	wininit.exe	392	324	5	80	0	0	2020-12-27 06:18:50 UTC+0000	
0xffffffff8003a31060	winlogon.exe	428	376	5	120	1	0	2020-12-27 06:18:50 UTC+0000	
0xffffffff80029e27c0	services.exe	488	392	18	239	0	0	2020-12-27 06:18:51 UTC+0000	
0xffffffff80029e9aa0	lsass.exe	496	392	8	597	0	0	2020-12-27 06:18:51 UTC+0000	
0xffffffff80029ea880	lsm.exe	504	392	10	154	0	0	2020-12-27 06:18:51 UTC+0000	
0xffffffff8003aa5520	svchost.exe	592	488	14	372	0	0	2020-12-27 06:18:54 UTC+0000	
0xffffffff8003ac1060	vm3dservice.ex	652	488	3	45	0	0	2020-12-27 06:18:54 UTC+0000	
0xffffffff8003acfb30	svchost.exe	692	488	10	303	0	0	2020-12-27 06:18:55 UTC+0000	
0xffffffff8003b04b30	svchost.exe	776	488	25	494	0	0	2020-12-27 06:18:55 UTC+0000	
0xffffffff8003b239e0	svchost.exe	816	488	26	447	0	0	2020-12-27 06:18:55 UTC+0000	
0xffffffff8003b2db30	svchost.exe	840	488	43	805	0	0	2020-12-27 06:18:55 UTC+0000	
0xffffffff8003b6e060	audiodg.exe	920	776	6	128	0	0	2020-12-27 06:18:56 UTC+0000	
0xffffffff8003b99b30	svchost.exe	1000	488	28	745	0	0	2020-12-27 06:18:57 UTC+0000	
0xffffffff8003bc5060	svchost.exe	376	488	20	389	0	0	2020-12-27 06:18:58 UTC+0000	
0xffffffff8003c35060	spoolsv.exe	1080	488	16	290	0	0	2020-12-27 06:19:00 UTC+0000	
0xffffffff8003c47160	svchost.exe	1108	488	24	318	0	0	2020-12-27 06:19:00 UTC+0000	
0xffffffff8003cbbb30	svchost.exe	1232	488	28	327	0	0	2020-12-27 06:19:01 UTC+0000	
0xffffffff8003ca8b30	VGAuthService.	1348	488	6	93	0	0	2020-12-27 06:19:02 UTC+0000	
0xffffffff8003db47d0	vmtoolsd.exe	1476	488	11	269	0	0	2020-12-27 06:19:05 UTC+0000	
0xffffffff8003d76740	dllhost.exe	1764	488	21	195	0	0	2020-12-27 06:19:09 UTC+0000	
0xffffffff8002e60740	dllhost.exe	1840	488	18	209	0	0	2020-12-27 06:19:09 UTC+0000	
0xffffffff8003e88890	msdtc.exe	1924	488	15	157	0	0	2020-12-27 06:19:10 UTC+0000	
0xffffffff8003e9a060	WmiPrvSE.exe	1224	592	9	184	0	0	2020-12-27 06:19:12 UTC+0000	
0xffffffff8003eea620	VSSVC.exe	880	488	6	123	0	0	2020-12-27 06:19:12 UTC+0000	
0xffffffff8003eb97c0	WmiPrvSE.exe	620	592	13	300	0	0	2020-12-27 06:19:29 UTC+0000	
0xffffffff8003fc2b30	WmiApSrv.exe	2040	488	7	121	0	0	2020-12-27 06:19:32 UTC+0000	
0xffffffff8004006470	taskhost.exe	2108	488	10	164	1	0	2020-12-27 06:19:47 UTC+0000	
0xffffffff800401cb30	userinit.exe	2156	428	3	46	1	0	2020-12-27 06:19:47 UTC+0000	
0xffffffff80040247d0	dwm.exe	2168	816	5	107	1	0	2020-12-27 06:19:47 UTC+0000	
0xffffffff8004042b30	explorer.exe	2220	2156	28	636	1	0	2020-12-27 06:19:47 UTC+0000	
0xffffffff80040fe790	vm3dservice.ex	2324	2220	2	49	1	0	2020-12-27 06:19:50 UTC+0000	
0xffffffff80040fd9e0	vmtoolsd.exe	2332	2220	9	158	1	0	2020-12-27 06:19:50 UTC+0000	
0xffffffff8003afe5e0	SearchIndexer.	2496	488	14	579	0	0	2020-12-27 06:19:55 UTC+0000	
0xffffffff80029d1b30	SearchProtocol	2616	2496	7	249	0	0	2020-12-27 06:19:58 UTC+0000	
0xffffffff80041c22e0	SearchFilterHo	2648	2496	5	83	0	0	2020-12-27 06:19:58 UTC+0000	
0xffffffff80041dbb30	wmpnetwk.exe	2692	488	19	397	0	0	2020-12-27 06:19:58 UTC+0000	
0xffffffff8004235b30	svchost.exe	2852	488	5	174	0	0	2020-12-27 06:20:00 UTC+0000	

File integrity check using Hash on OSForensics



File Hashing

Hash Sets

Verify/Create Hash

☒ File☐ Volume☐ Text

File

TOPSECRET.txt

...

Calculate

Hash Function

MD5

Secondary Hash Function

MD5

☐ Upper case output

Progress

Data Hashed

42 Bytes

Calculated Hash

bcd5abecd5bcae4570532d217330079e

MD5

Primary

bcd5abecd5bcae4570532d217330079e

MD5

Secondary

Comparison Hash

bcd5abecd5bcae4570532d217330079e



Hashes (primary) are equal

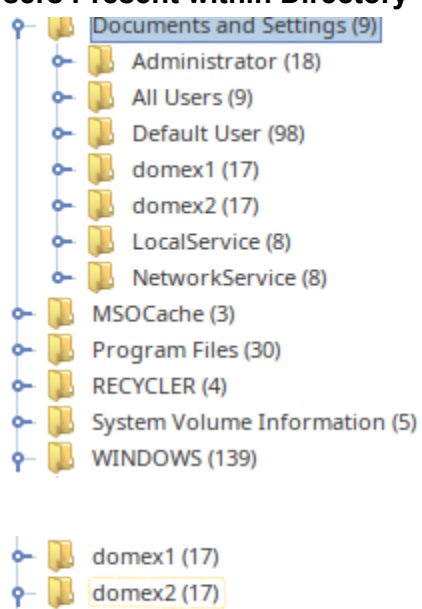
Add Result to Case...

Selected Hash Function Description

MD5 (Message-Digest algorithm 5) is an internet standard cryptographic hash function. MD5 has been found to not be collision resistant and therefore not suitable for many security applications.

Due to its popularity, MD5 is still used in many situations where security is not of high importance or for legacy purposes.

Users Present within Directory



An image present within the user's directory

[current folder]			2008-10-30 05:51:26 GST	2008-10-30 05:51:26 GST	2008-10-30 11:50:13 GST	2008-10-21 23:29:54 GST
[parent folder]			2008-10-21 23:29:59 GST	2008-10-30 05:51:18 GST	2008-10-30 11:50:13 GST	2008-10-21 23:29:51 GST
Desktop.ini			2008-10-21 23:29:59 GST	2008-10-21 23:29:59 GST	2008-10-30 06:42:18 GST	2008-10-21 23:29:54 GST
domexuser2.JPG			2008-10-29 20:21:16 GST	2008-10-30 05:59:10 GST	2008-10-30 05:51:26 GST	2008-10-29 20:21:16 GST
Sample Pictures.Ink			2008-10-21 23:29:55 GST	2008-10-21 23:29:55 GST	2008-10-30 05:51:24 GST	2008-10-21 23:29:55 GST
Thumbs.db			2008-10-30 05:51:28 GST	2008-10-30 05:51:28 GST	2008-10-30 05:51:26 GST	2008-10-30 05:51:26 GST
Thumbs.db:encryptable			2008-10-30 05:51:28 GST	2008-10-30 05:51:28 GST	2008-10-30 05:51:26 GST	2008-10-30 05:51:26 GST

Report Generation Templates

Generate Report

Select and Configure Report Modules

Report Modules:

- ☒ HTML Report
- ☐ Excel Report
- ☐ Files - Text
- ☐ Data Source Summary Report
- ☐ Save Tagged Hashes
- ☐ Extract Unique Words
- ☐ TSK Body File
- ☐ Google Earth KML
- ☐ CASE-UCO
- ☐ Portable Case

A report about results and tagged items in HTML format.

Header:

Footer:

< Back Next > Finish Cancel Help

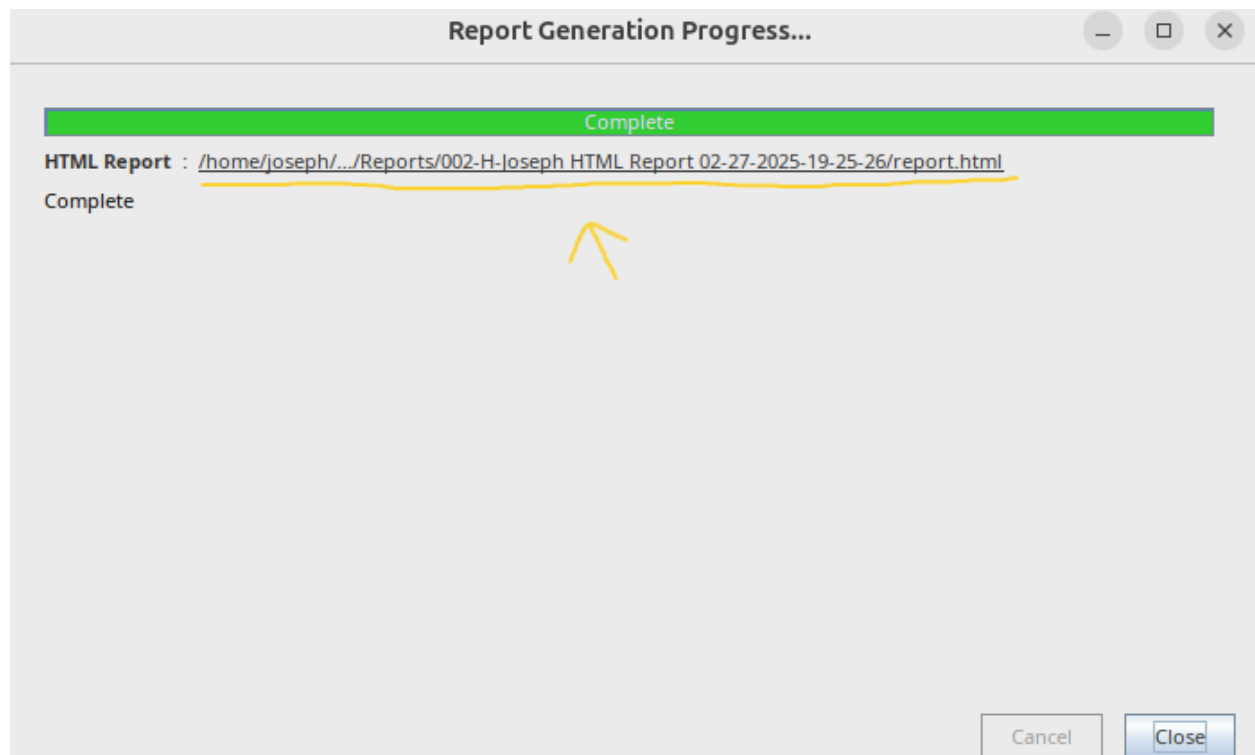
Report Generation with Tagged Items

The screenshot shows a 'Generate Report' dialog box with a 'Configure Report' tab. The dialog has a title bar with standard window controls. The main area is titled 'Configure Report' and contains the following elements:

- A label 'Select which data to report on:' followed by three radio buttons:
 - ☐ All Results
 - ☒ All Tagged Results
 - ☐ Specific Tagged Results
- A large rectangular list box containing one item: ☒ Notable Item (Notable).
- Two buttons to the right of the list box: 'Select All' and 'Deselect All'.
- A button labeled 'Choose Result Types...' located below the list box.

At the bottom of the dialog, there is a row of five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Report Generation Stored in Files



Stringed data from a file in Timeline

